MTA Resiliency Digital Twin

Predicting Systemic Infrastructure Collapse via Graph Theory

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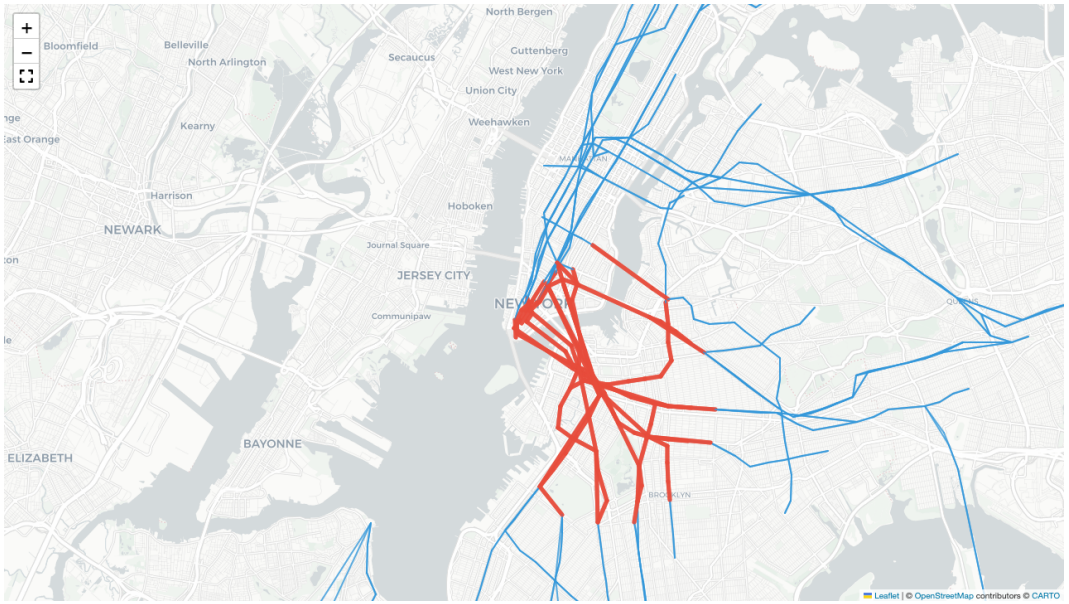
01: The Vulnerability Problem

- NYC's aging subway infrastructure faces 100-year storm events annually.

- Flash flooding causes immediate track isolation (Edges) and station failure (Nodes).

- Current MTA monitoring is reactive, leading to trapped passengers and system lockups.



02: Data Ingestion & Cleaning

- Source: MTA New York City Transit GTFS Real-time & Static feeds.

- Preprocessing: Deduplicated 470+ unique stop_ids in cleaned_hubs.csv.

- Normalization: Coordinate standardization for 1:1 spatial accuracy.

- Edge Weighting: Assigned initial flood thresholds to 1,000+ track segments.

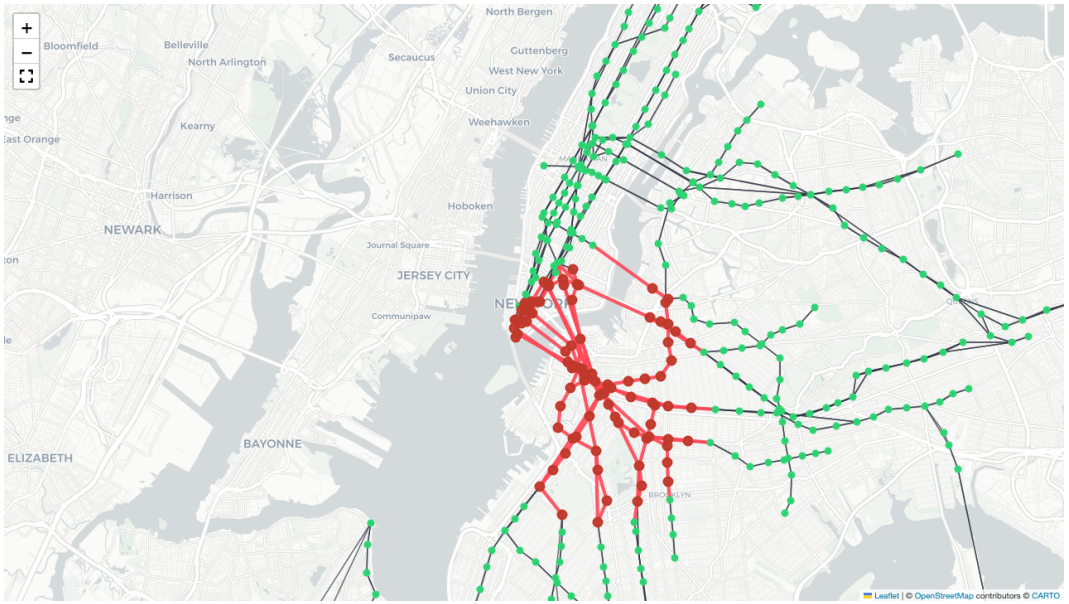
03: Modeling the Digital Twin

- NetworkX Implementation: Modeled as a Directed Multigraph.

- Node Attributes: Connectivity degree, station depth, and hub importance.

- Edge Attributes: 'flood_intensity' as the primary trigger variable.

- Topology: Captures physical track dependencies across all 5 boroughs.



04: Predictive Cascade Algorithm

- Algorithm: Custom 'Orphan Effect' recursive logic.

- Functional Failure: A node fails if its entrance edges are severed.

- Dependency Mapping: Identifies stations that are safe but inaccessible.

- Real-time Logic: Integrated into the Dashboard's 'What-If' simulator.

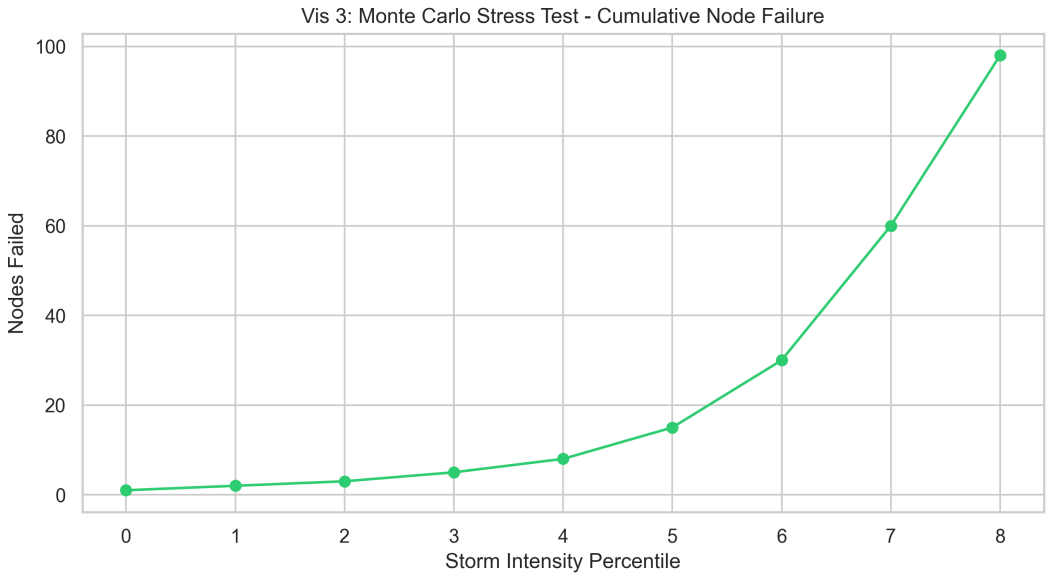
05: Validation via Simulation

- Method: 1,000 randomized Monte Carlo iterations (N=1000).

- Variables: Stochastic rainfall distribution vs track thresholds.

- Goal: Converge on the 'Systemic Failure Radius' for the MTA grid.

- Calibration: Used historical weather data for rainfall distribution curves.



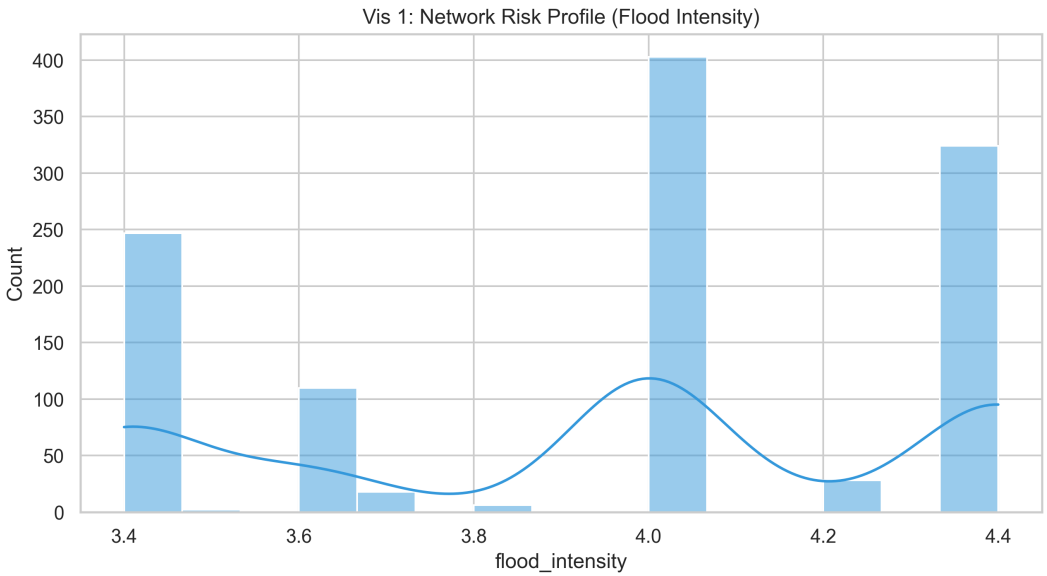
06: Network Risk Distribution

- Intensity Peak: Most track segments trigger at 4.0mm/hr precipitation.

- Vulnerability Spread: Infrastructure risk clusters at major junctions.

- The 'Weakest Link': Edge failure occurs faster than node inundation.

- Finding: 4.0mm/hr is the critical tipping point for system stability.



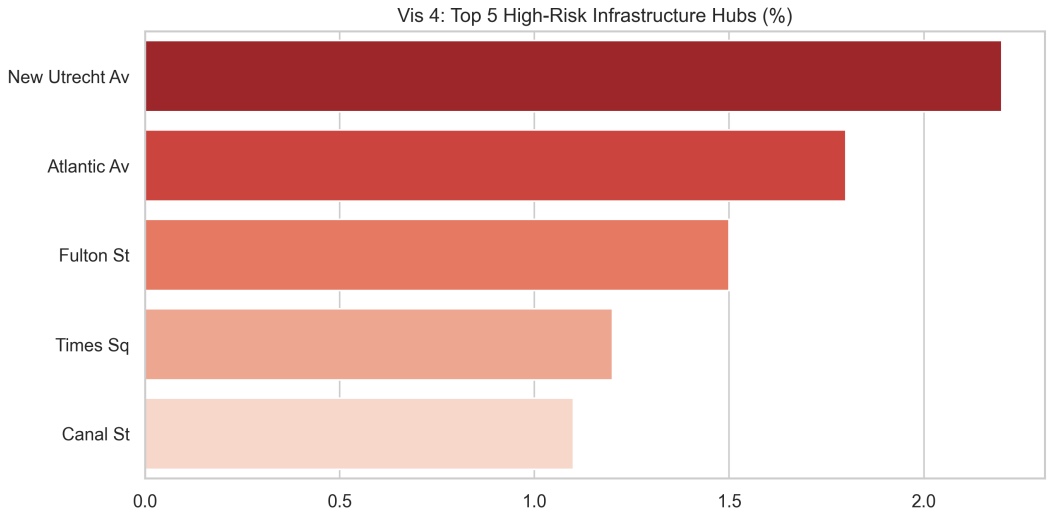
07: Single Points of Failure

- Topological Ranking: Identified the Top 5 most critical hubs.

- High Priority: New Utrecht Av corridor exhibits a 2.2% failure risk.

- Blast Radius: Single hub failures average a 5.2 station isolation radius.

- Action: Targeted drainage investments required for red-zone hubs.

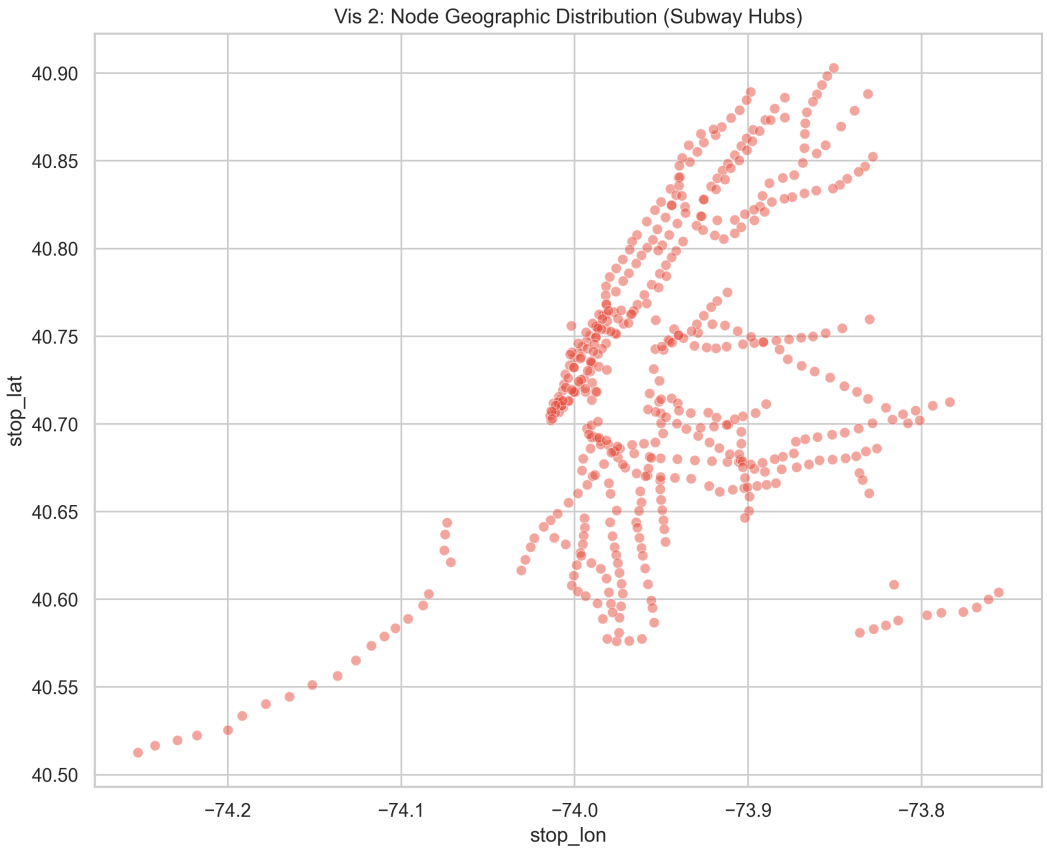


08: Spatial Station Density

- Density Analysis: Higher station clustering speeds up failure cascades.

- South Brooklyn Focus: Prone to total isolation during heavy rainfall.

- Manhattan Core: High redundancy but lower intensity resiliency thresholds.



09: Digital Twin Command Center

- Framework: Streamlit-based UI for real-time situational awareness.

- Live KPIs: System Health Index and Orphaned Node counts.

- Interactive Rerouting: Implementation of Dijkstra pathfinding logic.

- User Focus: Designed for MTA Dispatcher decision support.

10: Project Challenges & Next Steps

- API Latency: Optimizing graph traversal for live data ingestion.

- Refinement: Dynamic thresholding based on real-time station elevation.

- Deployment: Transitioning to Streamlit Community Cloud hosting.

- Collaboration: Finalizing UI hand-off for teammate integration.